

superconductivity in 1957. Bardeen suggested that whereas electrons usually repel each other, at very low temperatures they attract each other. Cooper said that the electrons worked together in what became known as Cooper pairs, and Schrieffer stated that the Cooper pairs moved in a coordinated way, as a single wave, so that the electrical resistance disappeared. In 1972, the three physicists were jointly awarded the Nobel Prize for their theory. Superconductors now have a wide range of applications, including use in powerful electromagnets in magnetic resonance (MRI) machines and maglev trains, which levitate above the tracks.

WILLIAM SHOCKLEY

American physicist and inventor William Shockley helped make the first transistor.



Shockley (1910–1989) had a background in radar research and solid state electronics, and was one of the first to work with silicon as a semiconductor. While at Bell Telephone Laboratories, he helped invent the transistor and then developed it further to produce the junction transistor. In 1955, he founded Shockley Semiconductor, one of the first electronics companies in what is now Silicon Valley.

BARDEEN AND BRATTAIN'S TRANSISTOR WAS 0.5 IN (1.25 CM) HIGH

PROVIDED THE FIRST SUCCESSFUL MODEL OF SUPER CONDUCTIVITY

Bardeen's transistor started a revolution in the electronics industry. It was favored for its size and reliability over vacuum tubes; most electronic devices would become transistor-based within 25 years.



1 OF JUST 4 PEOPLE TO HAVE WON THE NOBEL PRIZE TWICE

